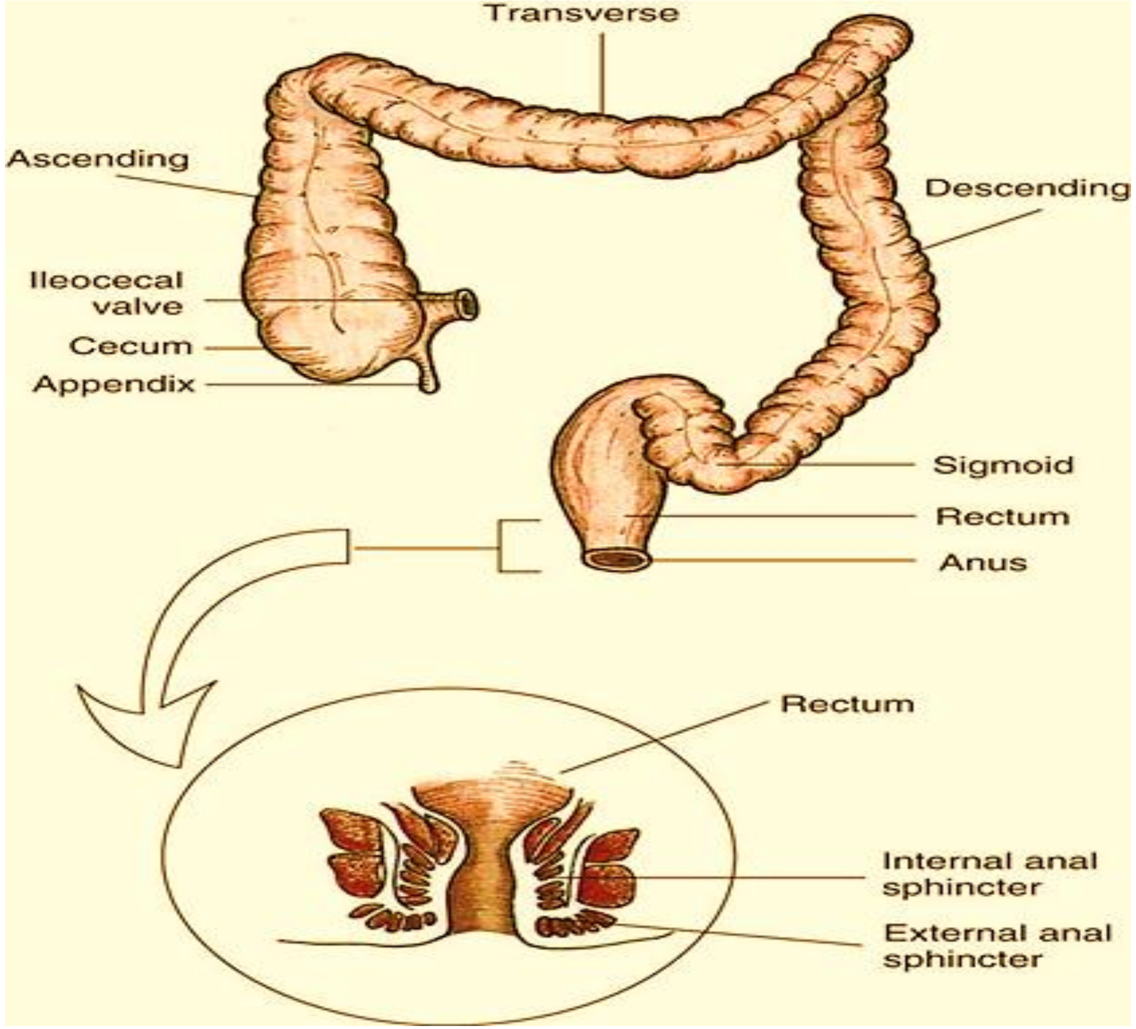
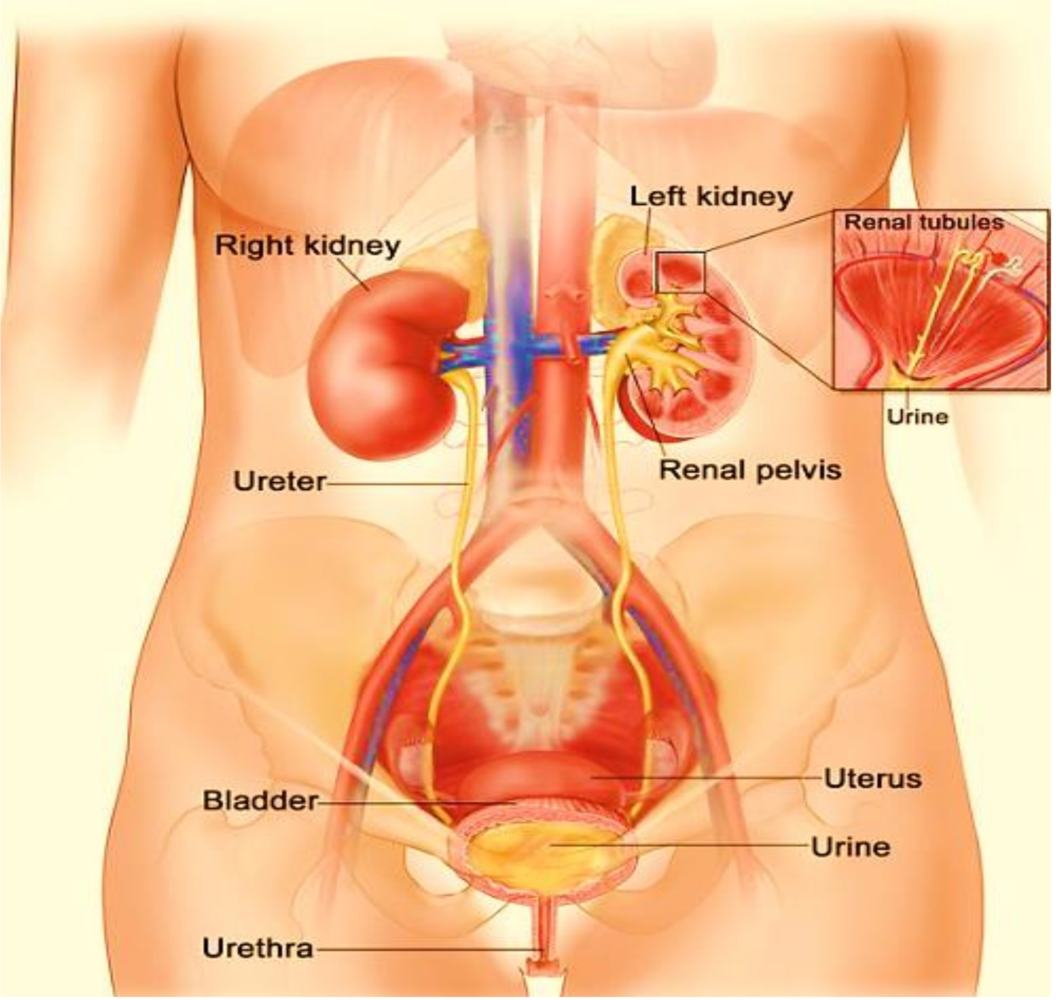
Fecal & Urinary elimination





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Outlines:

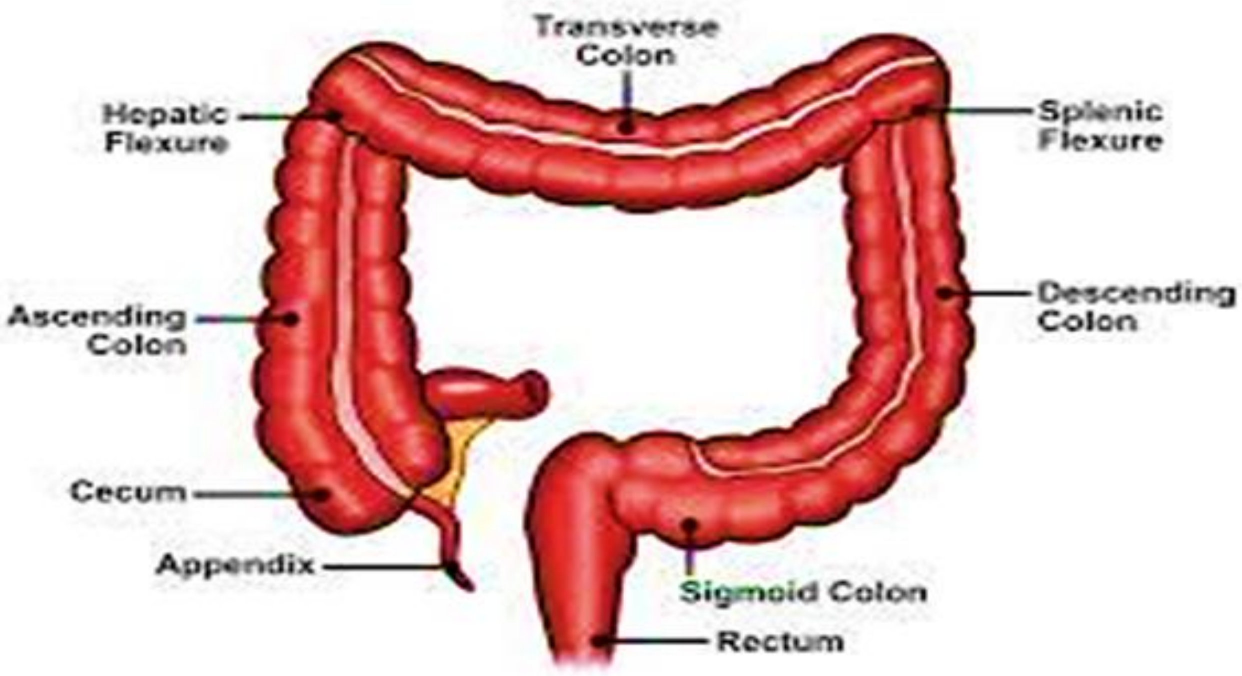
- Introduction
- Basic facts in relation to anatomy of the large intestine
- The act of defecation
- Factors influencing bowel elimination
- Common bowel elimination problems:

Constipation	Impaction	Diarrhea
Incontinence	Flatulence	Hemorrhoids

Bowel elimination

Introduction:

Elimination is essential to rid the body of wastes and materials in excess of bodily needs. Elimination process is necessary to maintain high level of wellness and even life itself and must continue during illness and health.



Basic facts in relation to anatomy of the large intestine (colon):

1. The large intestine is a tube leading from "' the end of small intestine to the external skin and is about 150 -180 cm in length. The ileocecal valve separates the small intestine from the large intestine. It opens in one direction; prevent the passage of material in the opposite direction

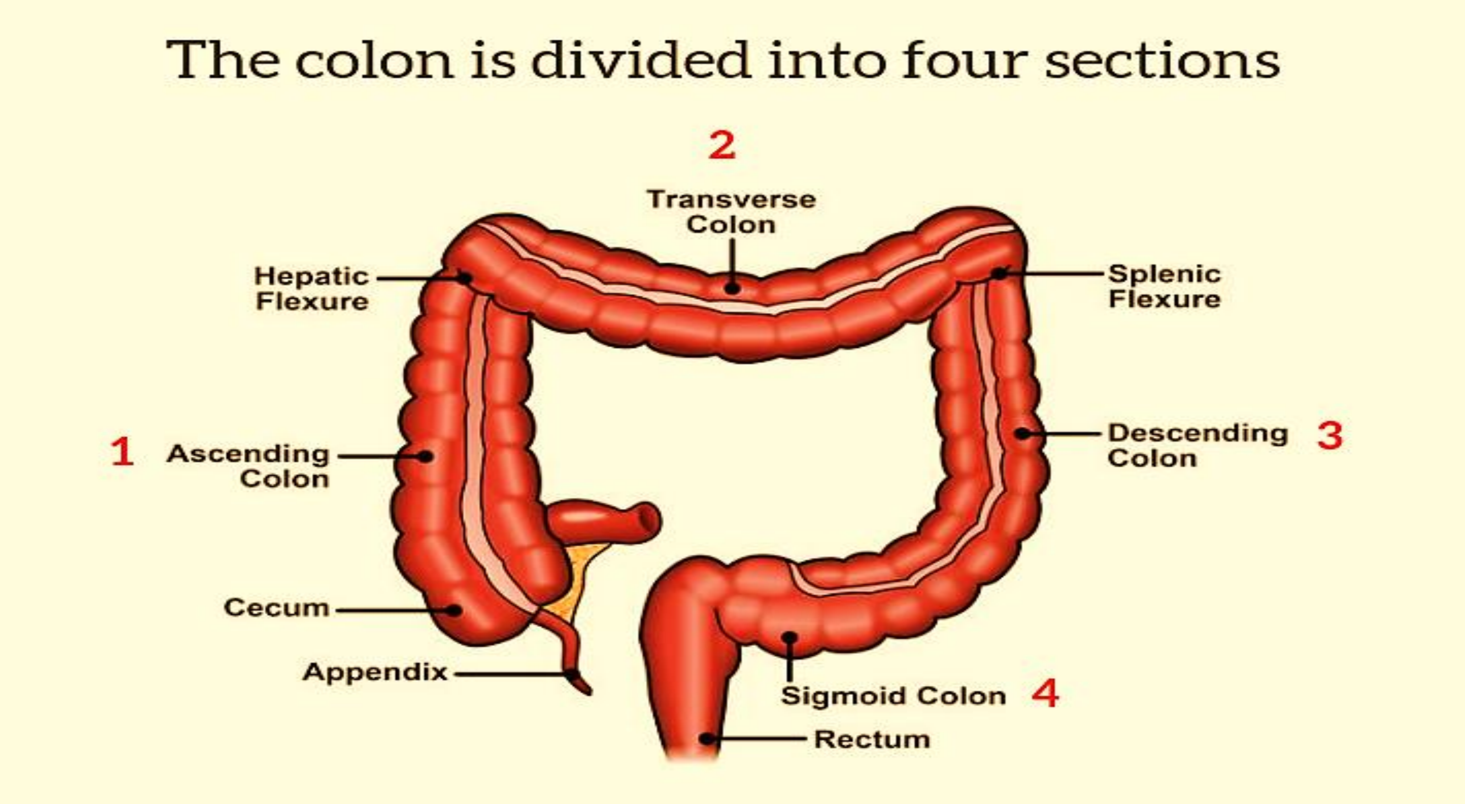
2. The large intestine is divided into:

- The cecum: lies at the beginning of the large intestine.
- The colon: lies between the cecum and the rectum and is divided into: The ascending colon goes up on the right side, the transverse colon crosses the abdomen and the descending colon goes down on the left side, The sigmoid flexure ends at the rectum.

Basic facts in relation to anatomy of the large intestine (colon):

- The rectum of the adult person is about 15-20 cm.

3. The anal canal is about 2.5 cm, and has two sphincters. The internal sphincter and the external sphincter at the anus, the external sphincter has striated muscles and is under voluntary control.



Defecation

Is an evacuation of the intestines and is often referred to as a bowel movement. When a certain amount of fecal matter accumulates in the rectum it becomes rises.

Factors influencing bowel elimination

1. Age
2. Diet
3. Fluid intake
4. Physical activity
5. Psychological factors
6. Personal habits
7. Position during defecation
8. Pain
9. Pregnancy
10. Surgery and anesthesia

Factors influencing bowel elimination

1. Age

- **An infant** has a small stomach capacity and less secretion of digestive enzymes. Food passes quickly through an infant's intestinal tract because of rapid peristalsis. The infant is unable to control defecation because of a lack of neuromuscular development. This neuromuscular development usually does not **take place until 2 to 3 years of age.**
- **Older adults** often have difficulty controlling bowel evacuation and are at risk for incontinence. Older adults sometimes develop irregular bowel movements and an increased risk for constipation.

Factors influencing bowel elimination

2. Diet

- Ingestion of a high-fiber diet improves the likelihood of a normal elimination pattern if other factors are normal. Diets high in vegetables and fruits have been linked to decreased risk of colorectal cancer.

- **Gas-producing foods** such as onions and beans also stimulate peristalsis. Some spicy foods increase peristalsis but also cause indigestion and watery stools.

- **Food intolerance** is not an allergy but rather a particular food that causes the body distress within a few hours of ingestion. The result is diarrhea, cramps, or flatulence. For example, people who drink cow's milk and have these symptoms are not allergic to milk but lack the enzyme needed to digest the milk sugar lactase and therefore are lactose intolerant.

Factors influencing bowel elimination

3. Fluid Intake:

- An adult need to drink at least 1100 to 1400 mL of fluid daily. An increase in fluid intake with the use of fruit juices softens stool and increases peristalsis. Poor fluid intake increases the risk of constipation.

4. Physical Activity:

- Physical activity promotes peristalsis, whereas immobilization depresses it.

Factors influencing bowel elimination

5. Psychological Factors

- During emotional stress the peristalsis is increased causing diarrhea and gaseous distention.
- Several diseases of the gastrointestinal tract are associated with stress, including ulcerative colitis, gastric and duodenal ulcers.
- In depression state the autonomic nervous system slows impulses, peristalsis decreases, resulting in constipation.

Factors influencing bowel elimination

6. Personal Habits

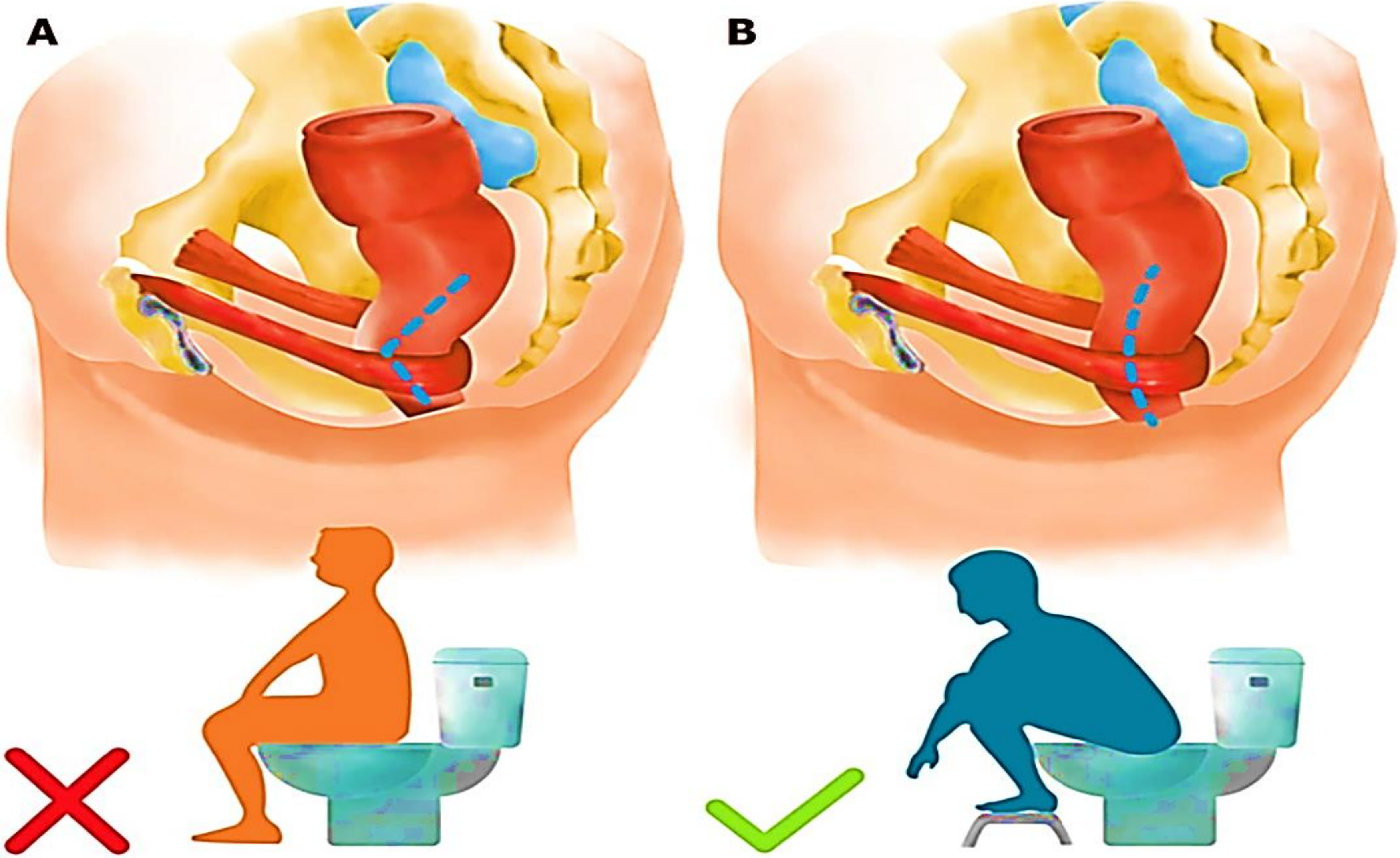
- Most people benefit from being able to use their own toilet facilities at a time that is most effective and convenient for them.

- Chronically ill and hospitalized patients are not always able to maintain privacy during defecation. The sights, sounds, and odors associated with sharing toilet facilities or using bedpans are causes patients to ignore the urge to defecate causing constipation and discomfort.

7. Position during defecation

- Squatting is the normal position during defecation. For the patient immobilized in bed, defecation is often difficult.

Squatting position



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Factors influencing bowel elimination

8. Pain: Normally the act of defecation is painless. the patient often suppresses the urge to defecate to avoid pain in conditions such as hemorrhoids, rectal surgery, and abdominal surgery that contribute to develop constipation.

9. Pregnancy: As pregnancy advances, the size of the fetus increases, and pressure is exerted on the rectum impairs passage of feces leads to constipation. A pregnant woman's frequent straining during defecation or delivery results in formation of permanent hemorrhoids.

10. Surgery and Anesthesia: General anesthetic agents used during surgery cause temporary cessation of peristalsis. Any surgery that involves direct manipulation of the bowel temporarily stops peristalsis. This condition, called **paralytic ileus,** usually lasts about 24 to 48 hours.

Common Bowel Elimination Problems

1. Constipation
2. Fecal impaction
3. Diarrhea
4. Incontinence
5. Flatulence
6. Hemorrhoids

1. Constipation

Definition: The passage of unusually dry, hard stools produced by undue delay in the passage of feces.

Causes of constipation

-Poor elimination habits. If the desire for defecation is ignored repeatedly, the feces become hard and dry because of increased water absorption.

- Lack of sufficient roughage or bulk in diet.

-Lack of enough fluid intakes.

-Lack of essential vitamins such as vitamin B.

-Lack of exercise

Assessment of patient with constipation:

- Passage of hard stools associated with a decrease in the usual frequency of defecation.
- Abdominal distension (the abdomen feels hard upon palpation) caused by accumulation of fecal matter as well as gases.
- Complaints of tenesmus (frequent painful straining in attempts to defecate).
- General symptoms: e.g. headache, malaise, anorexia, and bad breath.

Nursing management of constipation

1. Provide adequate fluid intake 500 – 2000 cc/day.

2. Provide a well-balanced diet with enough roughage from fruits and vegetables and vitamins.

3. Encourage regularity of time for defecation and prompt response to the desire of defecation.

4. Provide adequate time for complete

5. Provide privacy for patients to promote relaxation.

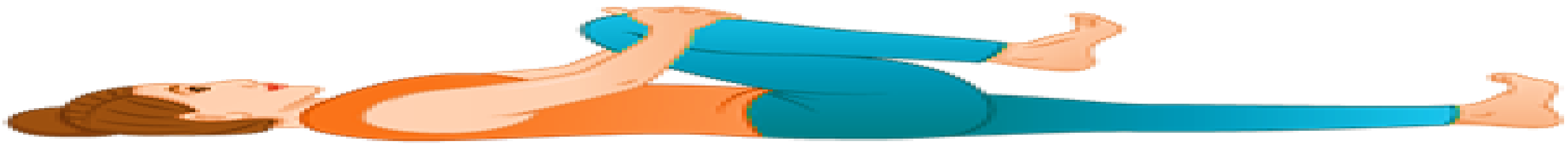
6. Provide posture (position) as close to normal as possible.

7. Consider the patient's habit in relation to defecation.

8. Provide physical exercises especially for abdominal muscles

- Lie supine; tighten abdominal muscles as though pushing them to the floor. Hold muscles tight to count of three; relax. Repeat 5 to 10 times as tolerated.

- Flex & contract thigh muscles by raising one knee slowly toward the chest. Repeat for each leg at least 5 times & increase frequency as tolerated.



9. Avoiding medications such as opioids that increase constipation

10. An enema is the instillation of a solution into the rectum and sigmoid colon. The primary reason for an enema is to promote defecation by stimulating peristalsis.

Prevention of constipation

- Encourage exercise as walking.
- Avoid excessive emotional stress.
- Establish regularity of meals and defecation time.
- Discourage unnecessary use of laxatives.
- Intake of proper diet containing enough vegetables and vitamins.
- Intake of sufficient fluids per day.

2. Fecal impaction

Definition: A prolonged retention or an accumulation of fecal material which forms a hardened mass in the rectum, it may be of sufficient size to prevent the passage of normal stools.

Causes

- Prolonged constipation and poor habits of defecation.
- Prolonged bed rest, vary in paralyzed or unconscious patients.
- Prolonged use of anti-diarrheas drugs.
- Following administration of Barium for x-ray examination of the G.I.T.

Fecal Impaction
For Information,
Visit: www.epainassist.com
Descending Colon
Small Intestine
More Stool Form and Back to Colon
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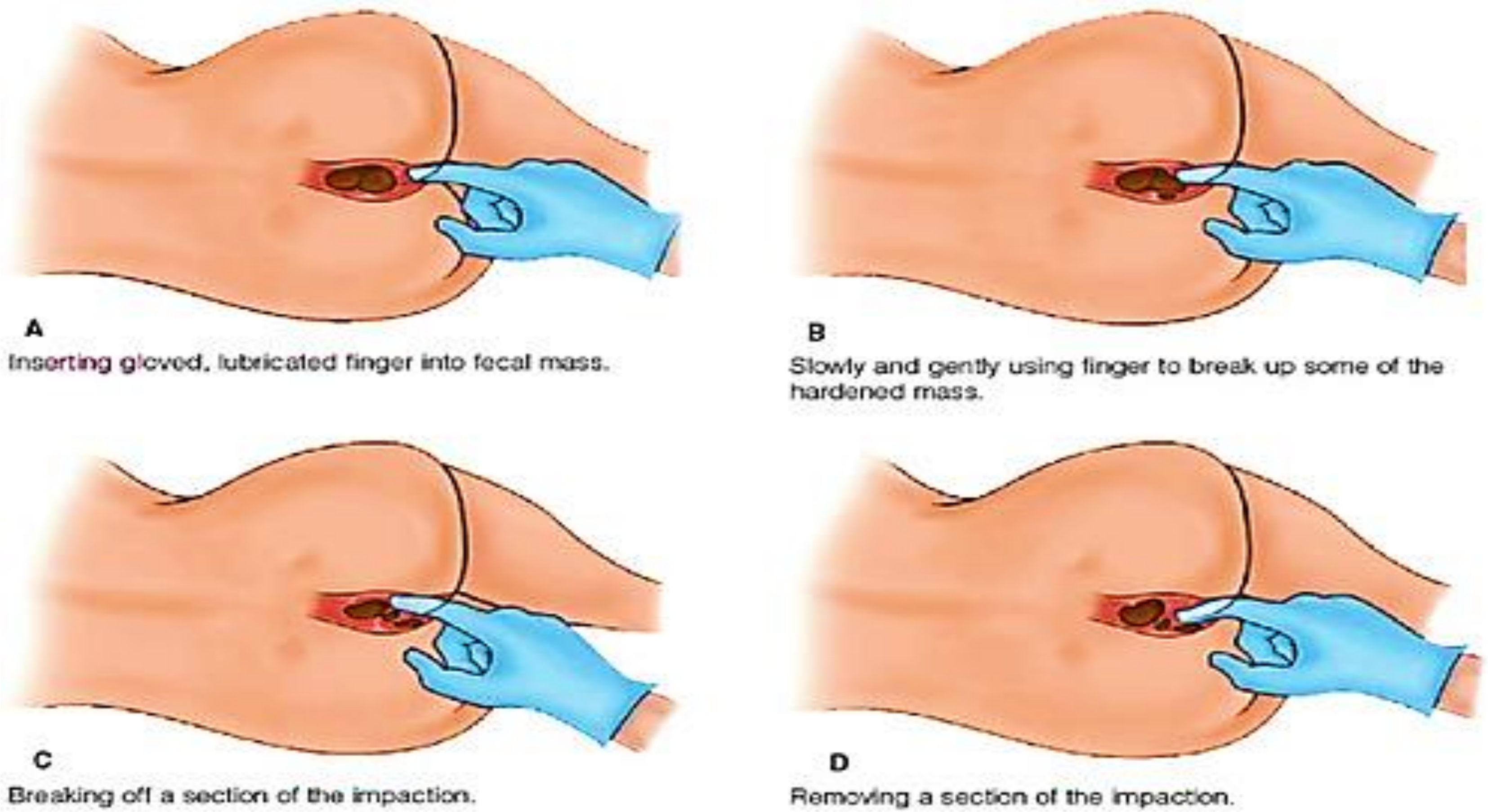
Signs and symptoms of fecal impaction

- Distended abdomen (hand upon palpation and feels rigid).
- Rectal pain due to pressure of the fecal mass.
- Passage of small amount of liquid stool due to mechanical irritation of the rectum.

Nursing management of fecal impaction

- Administration of mineral oil by mouth especially in cases of prolonged constipation for regulation of habits.
- Oil retention enema followed by cleansing enema.

- Digital manipulation of the fecal mass should be under physician order or supervision because it can stimulate vague nerve in the rectal wall which can slow patient's heart leading to cardiac arrhythmia, so observe patient's pulse rate, facial pallor and diaphoresis during manipulation.



Prevention of fecal impaction:

- Careful observation of the patient's bowel movements in terms of amount, consistency, and frequency.
- Prevention of constipation.
- Special attention to patients who received barium for x-ray of the G.I.T

3. Diarrhea

Definition: Is the passage of loose, watery stool and an increase in the frequency of bowel movements, diarrhea may or may not be accompanies by abdominal cramping.

Causes:

- Patients receiving enteral nutrition are also at risk for diarrhea.
- Food allergies and intolerances increase peristalsis
- Surgeries or diagnostic testing of the lower GI tract
- Communicable food-borne pathogens

Signs and symptoms of diarrhea

- Generalized abdominal pain.
- Feeling of urgency in the need to defecate.
- Complaints of tenesmus and may pass a small watery discharge.
- Increase in the frequency in the number of stool (stool is watery in nature).
- Signs and symptoms of dehydration may occur if diarrhea very severs or over a long time such as: poor skin turgor, thirst, and acute weight loss.
- General weakness and general malaise.

Nursing management of diarrhea

1. Drink at least eight glasses of water per day to prevent dehydration.

2. Ingest foods with sodium and potassium. Most foods contain sodium. Potassium is found in meats, and many vegetables and fruits, especially tomatoes, potatoes, bananas, peaches, and apricots.

3. Increase foods containing soluble fiber, such as oatmeal, skinless fruits and potatoes. Limit fatty foods.

4. Avoid alcohol and beverages with caffeine, which aggravate the problem.

5. Limit foods containing insoluble fiber, such as whole-grain breads and cereals, and raw fruits and vegetables. Thoroughly clean and dry the perineal area after passing stool to prevent skin irritation and breakdown.

6. When diarrhea has stopped, re-establish normal bowel flora by taking fermented dairy products, such as yoghurt or buttermilk.

4. Incontinence

Definition: is the inability to control passage of feces and gas from the anus.

Causes

- Organic diseases causing weakness of the anal sphincter.
- Impingement in the nerve supply to the anal sphincter.

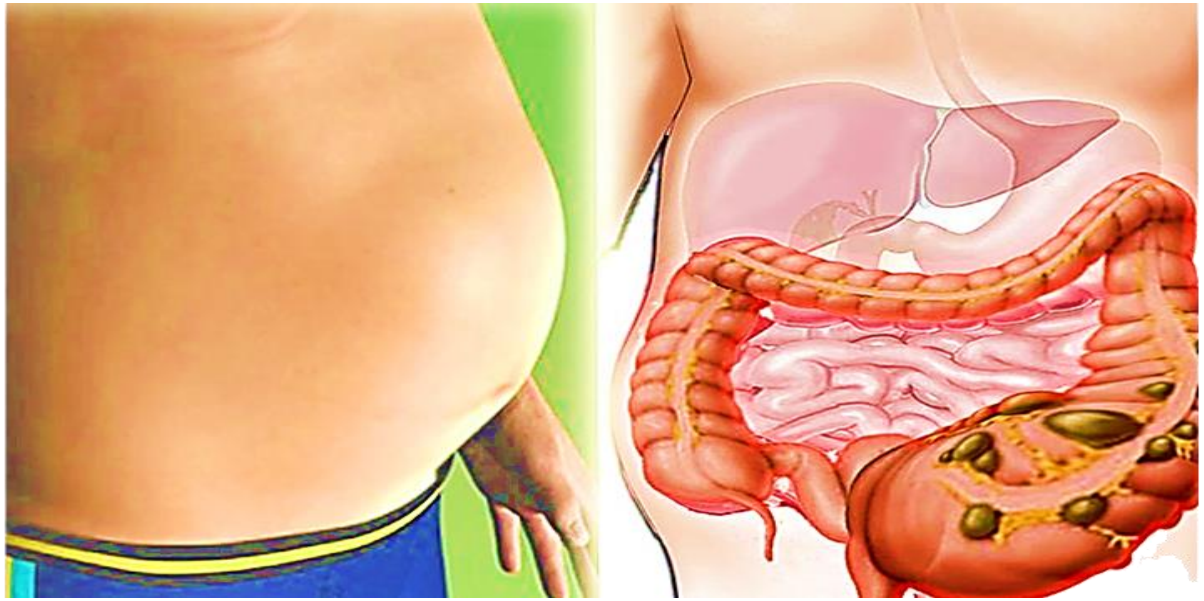
  (i.e. relaxed external sphincter).

Nursing management of Incontinence:

- Special nursing care to prevent bad odor, skin irritation and bed sores.
- Patient's clothing and bedding should be changed whenever necessary.

5. Flatulence

As gas accumulates in the lumen of the intestines, the bowel wall stretches and distends **(flatulence).** It is a common cause of abdominal fullness, pain, and cramping. Normally intestinal gas escapes through the mouth (belching) or the anus (passing of flatus). However, flatulence causes abdominal distention and severe, sharp pain if intestinal motility is reduced because of opiates, general anesthetics, abdominal surgery, or immobilization.



6. Hemorrhoids

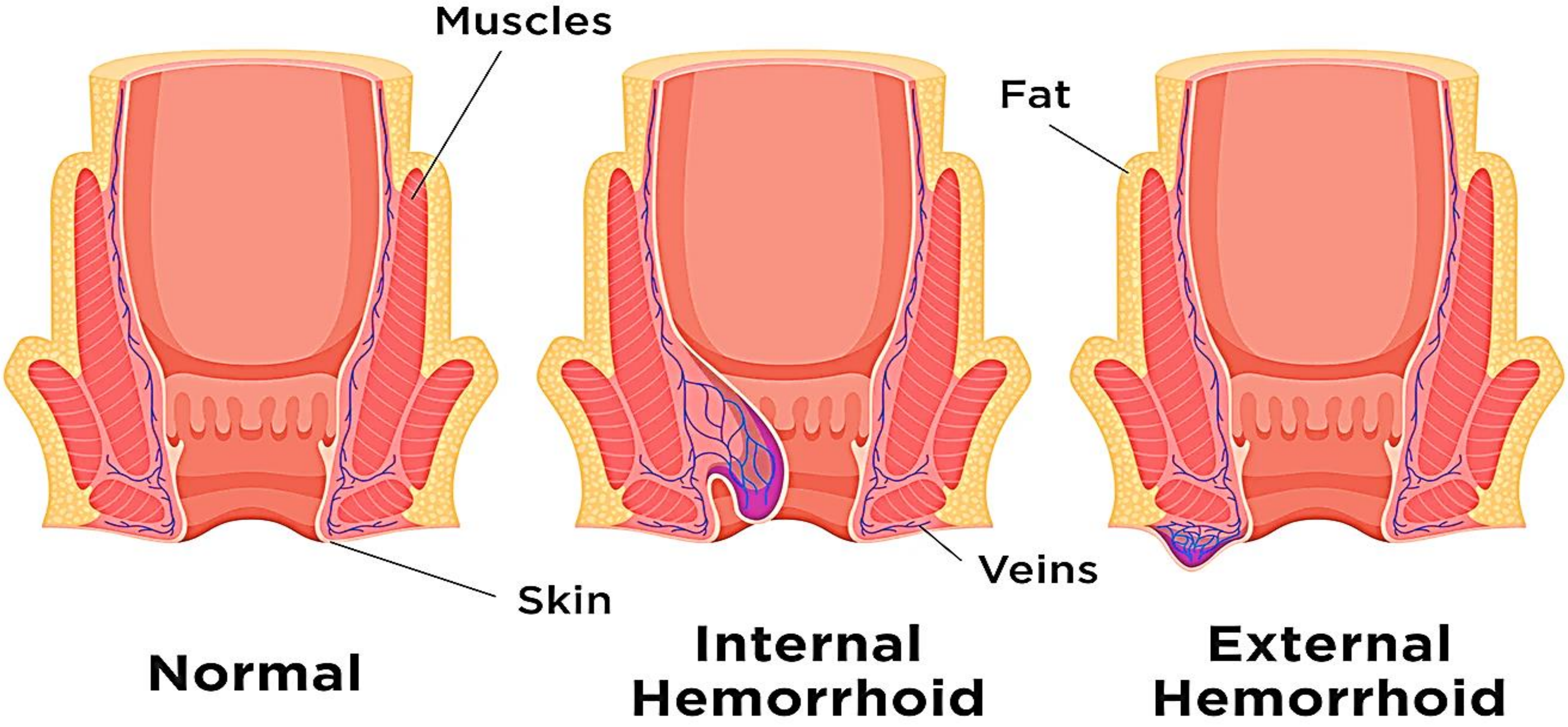
- Hemorrhoids are dilated, engorged veins in the lining of the rectum.

- They are either external or internal.

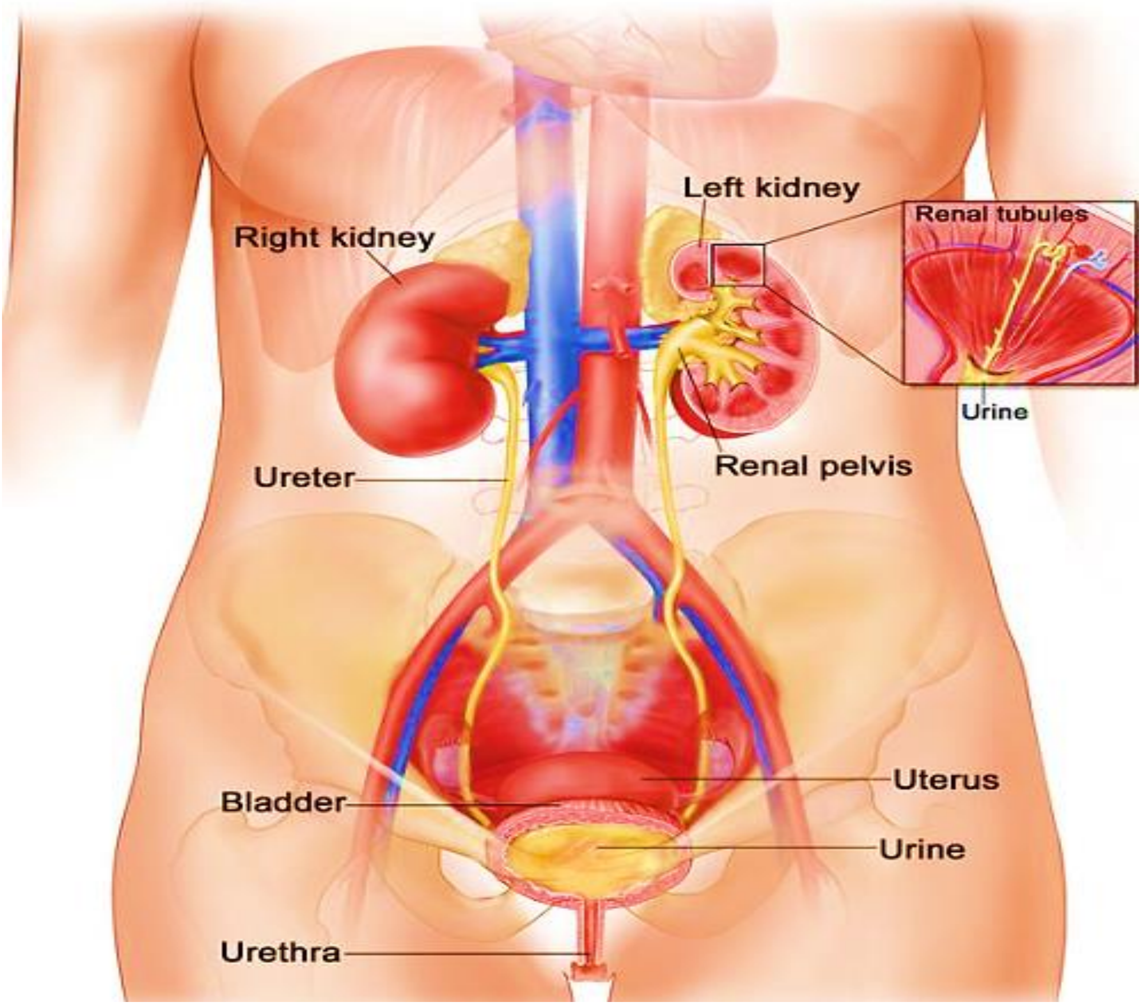
A. **External hemorrhoids** are clearly visible as protrusions of skin. If the underlying vein is hardened, there is usually a purplish discoloration (thrombosis). This causes increased pain and often needs to be excised.

B. **Internal hemorrhoids** have an outer mucous membrane. Increased venous pressure from straining at defecation, pregnancy, heart failure, and chronic liver disease causes hemorrhoids.

Hemorrhoid Types



Urinary elimination



Outlines:

1. **Introduction**
2. **Physiology of urinary elimination**
3. **Factors influencing urination.**
4. **Alterations in urinary elimination.**
5. **Common urinary elimination problems**
6. **Nursing management**

Introduction

Elimination from the urinary tract is usually taken for granted. A person's urinary habits depend on social culture, personal habits, and physical abilities.

Urinary elimination is essential to health, and voiding can be postponed for only so long before the urge normally becomes too great to control.

Urinary elimination depends on the effective functioning of the upper urinary tract's kidneys and ureters and the lower urinary tract's urinary bladder, urethra, and pelvic floor.

urination

Micturition, voiding, and urination all refer to the process of emptying the urinary bladder.

Urine collects in the bladder until pressure stimulates special sensory nerve endings in the bladder wall called stretch receptors. This occurs when the adult bladder contains between 250 and 450mL of urine.

In children, a considerably smaller volume, 50 to 200 mL, stimulates these nerves.

Factors Influencing Urination

Numerous factors affect the volume and characteristics of the urine produced and the manner in which it is excreted, these include the following:-

- **Developmental Factors**
- **Psychosocial Factors**
- **Fluid and Food Intake**
- **Medications**
- **Muscle Tone**
- **Pathologic Conditions**
- **Surgical and Diagnostic Procedures**

1. Developmental Factors

Stage	Variations
Fetuses	- The **fetal kidney begins to excrete urine between the 11th and 12th week of development**.
Infants	- Because of neuromuscular immaturity, **voluntary urinary control is absent.**
Children	- Kidney function **reaches maturity between the first and second year of life**. - Between 18 and 24 months of age, the child starts to recognize bladder fullness and is able to hold urine beyond the urge to void. At approximately 2 1/2 to 3 years of age, the child can perceive bladder fullness, hold urine after the urge to void, and communicate the need to urinate. - Full urinary control usually occurs at age 4 or 5 years; daytime control is usually achieved by age 3 years.
Adults	- **The kidneys reach maximum size between 35 and 40 years of age.** - After 50 years, the kidneys begin to diminish in size and function.
Older Adults	- An estimated 30% of nephrons are lost by age 80. - Renal blood flow decreases because of vascular changes and a decrease in cardiac output. - The ability to concentrate urine declines. Bladder muscle tone diminishes, causing increased frequency of urination and nocturia (awakening to urinate at night). - Diminished bladder muscle tone and contractibility may lead to residual urine in the bladder after voiding, increasing the risk of bacterial growth and infection. - Urinary incontinence may occur due to mobility problems or neurologic impairments.

2. Psychosocial Factors

- Urination is affected by a set of conditions that helps stimulate the micturition reflex. These are including privacy, normal position, sufficient time, and occasionally running water.
- Anxiety and stress lead to inhibition of voiding due to non-relaxation of abdominal and perineal muscles and contraction of the external urethral sphincter.
- People also may voluntarily suppress urination because of perceived time pressures; for example, nurses often ignore the urge to void until they are able to take a break. This behavior can increase the risk of UTIs.

3. Fluid and Food Intake

- The healthy body maintains a balance between the amount of fluid ingested and the amount of fluid eliminated. When the amount of fluid intake increases, therefore, the output normally increases.
- Certain fluids, such as alcohol, increase fluid output by inhibiting the production of antidiuretic hormone.
- Fluids that contain caffeine (e.g., coffee, tea, and cola drinks) also increase urine production.
- Food and fluids high in sodium can cause fluid retention because water is retained to maintain the normal concentration of electrolytes.
- Some foods and fluids can change the color of urine. For example, beets can cause urine to appear red; foods containing carotene can cause the urine to appear yellower than usual.

4. Medications

- **Diuretics** increase urine formation by preventing the re-absorption of water and electrolytes from the tubules of the kidney into the bloodstream.

- Some medications may alter the color of the urine.

5. Muscle Tone

- Good muscle tone is important to maintain the stretch and contractility of the detrusor muscle so the bladder can fill adequately and empty completely.

- Clients who require a retention catheter for a long period may have poor bladder muscle tone because continuous drainage of urine prevents the bladder from filling and emptying normally.

- Pelvic muscle tone also contributes to the ability to store and empty urine.

6. Pathologic Conditions

- Pre renal: Decreased blood flow to and through the kidney e.g. hemorrhage.

- Renal: Disease conditions of the renal tissue e.g. glomerulonephritis

- Post renal: Obstruction in the lower urinary tract that prevents urine flow from the kidney e.g. a urinary stone or Hypertrophy of the prostate gland that affect older men.

- Diseases that change nerve functions (neuropathies) as Diabetes mellitus can lead to loss of bladder tone, reduced sensation of bladder fullness and inability to inhibit bladder contractions.

- Diabetes insipidus increase urine formation due to a lack of the hormone vasopressin (antidiuretic hormone).

- Heart and circulatory disorders such as heart failure, shock, or hypertension can affect blood flow to the kidneys, interfering with urine production. If abnormal amounts of fluid are lost through another route (e.g., vomiting or high fever), the kidneys retain water and urinary output falls.

7. Surgical and Diagnostic Procedures

- The urethra may swell following a cystoscopy.
- Surgical procedures on any part of the urinary tract may result in some postoperative bleeding; as a result, the urine may be red or pink tinged for a time.
- Spinal anesthetics can affect the passage of urine because they decrease the client's awareness of the need to void.
- Surgery on structures adjacent to the urinary tract (e.g., the uterus) can also affect voiding because of swelling in the lower abdomen.

Common Types of Urinary Alterations

Symptoms	Descriptions	Causes or associated factors
Urgency	Feeling of need to void immediately	Full bladder, bladder irritation or inflammation from infection, overactive bladder, psychological stress
Dysuria	Painful or difficult urination	Bladder inflammation, trauma or inflammation of urethral sphincter.
Frequency	Voiding at frequent intervals (less than 2 hours)	Increased fluid intake, bladder inflammation, increased pressure on bladder (pregnancy), diuretic therapy
Hesitancy	Difficulty initiating urination	Prostate enlargement, anxiety, urethral edema
Polyuria	Voiding large amounts of urine	Excess fluid intake, diabetes mellitus, diabetes insipidus or kidney disease, use of diuretics, ingestion of fluids containing caffeine or alcohol
Oliguria	Diminished urinary output relative to intake (usually 400 mL/24 hr)	Dehydration, renal failure, urinary tract infection, increased Antidiuretic hormone secretion, heart failure
Nocturia	Voiding one or more times at night	Excessive fluid intake before bed (especially coffee or alcohol), renal disease, aging process, prostate enlargement
Dribbling	Leakage of urine despite voluntary control of urination	Stress incontinence, overflow from urinary retention (e.g., from benign prostatic hyperplasia)
Hematuria	Blood in urine	Neoplasms of kidney or bladder, glomerular disease, infection of kidney or bladder, trauma to urinary structures, calculi, bleeding disorders

Common Types of Urinary Alterations

Enuresis	involuntary urination in children beyond the age when voluntary bladder control is normally acquired, usually 4 or 5 years of age.	Family history of enuresis, difficult access to toilet facilities, home stresses
Incontinence	involuntary leakage of urine or loss of bladder control	Bladder inflammation, cerebrovascular accident (CVA; stroke), spinal cord injury, or other disease Difficulties in independent toileting (mobility impairment), cognitive impairment, recent anesthesia, recent perineal surgery

Common urinary elimination problems

1. **Urinary incontinence (UI)** or involuntary leakage of urine or loss of bladder control is a health symptom, not a disease. It is only normal in infants.

2. **Urinary Retention:** When emptying of the bladder is impaired, urine accumulates and the bladder becomes over-distended that causes poor contractility of the detrusor muscle and further impairing urination.

- Clients with urinary retention may experience overflow incontinence, eliminating 25 to 50 mL of urine at frequent intervals. The bladder is firm and distended on palpation.

Common causes of urinary retention include:

- prostatic (enlargement), surgery and some medications.
- **Acute urinary retention** is the most common complication in the first 2to 4 hours postoperatively.
- **Causes of chronic urinary retention** can include paraplegia, quadriplegia, multiple sclerosis, and urethral or perineal trauma

Urinary Tract Infections (UTI)

- **UTI is the most common health care acquired infection**; 80% of these infections result from the use of an **indwelling urethral catheter**.

- The patient's own colonic flora, including Escherichia coli, remains the most common causative pathogen. Microorganisms commonly enter the urinary tract through the ascending urethral route.

- Women are more susceptible to infection because of a short urethra and the proximity of the anus to the urethral meatus. In men the length of the urethra reduce the susceptibility to urinary tract infection.

- Poor perineal hygiene is another cause of urinary tract infection in women. Inadequate hand washing, failure to wipe from front to back after voiding or defecating, and frequent sexual intercourse predispose women to infection.

Signs and symptoms of urinary tract infection:

- Patients with lower urinary tract infections have pain or burning during urination (dysuria) as urine flows over inflamed tissues.
- Fever, chills, nausea, vomiting, and malaise develop as an infection worsens.
- An irritated bladder (cystitis) causes a frequent and urgent sensation of the need to void.
- Irritation to bladder and urethral mucosa results in blood-tinged urine (hematuria).
- The urine appears concentrated and cloudy because of the presence of white blood cells or bacteria.
- If infection spreads to the upper urinary tract (kidneys) causing pyelonephritis, flank pain, tenderness, fever, and chills are common.

Nursing management for urinary elimination problems

Nursing assessment

1. **Patient History:** The Nurse determine the patient's normal voiding pattern and frequency , appearance of the urine and any recent changes, any past or current problems with urination, the presence of an ostomy and factors influencing the elimination pattern.

2. **Physical assessment:** It includes assessment of the kidney, bladder, urethral meatus, integrity of skin surrounding the meatus.

3. **Assessing urine:** Normal urine consists of 96% water and 4% solutes. Organic solutes include urea, ammonia, creatinine and uric acid. Urea is the chief organic solute. Inorganic solutes include sodium, chloride, potassium, sulphate, magnesium, and phosphorus. Sodium chloride is the most abundant inorganic salt.

4. **Urinalysis** is an important diagnostic test used to assess patients for conditions such as urinary tract infections and renal calculi.

5. Measuring Residual urine:

- Post void residual (PVR) (urine remaining in the bladder following voiding) is normally 50 to 100 ml.

Diagnostic Tests for urinary elimination disorders:

Blood levels of two metabolically produced substances, urea and creatinine, are routinely used to evaluate renal function. Other tests related to urinary functions such as collecting urine specimens, measuring specific gravity.

Nursing diagnosis: The following nursing diagnosis that are related to urinary elimination problems include:

-Impaired Urinary Elimination (urinary retention or incontinence) -Body image disturbance

-Self-care deficit -Risk for impaired skin integrity

-Risk for infection -knowledge deficit

-Situational low self-esteem or social isolation

-Risk for deficient fluid volume or excess fluid volume

Nursing planning: Overall goals for patients with urinary elimination problems may include the following:

- Maintain or restore a normal voiding pattern.
- Regain normal urine output
- Prevent associated risks such as infection, skin breakdown, fluid and electrolyte imbalance, and lowered self-esteem.
- Perform toilet activities independently with or without assistive devices.

Nursing implementation

A. Patient Education

- learn about symptoms of urinary alterations so they can initiate early preventive health care.

- Learn proper perineal hygiene and hand washing to reduce the risks for infection.

B. Promoting Normal Micturition

1-Stimulating Micturition Reflex

- **Assuming normal position for voiding**

- A woman is better able to void in a squatting or sitting position. If the patient is unable to use toilet facilities, position him or her in a squatting position on a bedpan or bedside commode.

- A man voids more easily in the standing position. If the man cannot reach toilet facilities, have him stand at the bedside and void into a urinal.

- **sensory stimuli**

- The sound of running water helps many patients void through the power of suggestion.

- Stroking the inner aspect of the thigh stimulates sensory nerves and promotes the micturition reflex.

- Pouring warm water over the patient’s perineum and create the sensation to urinate.

2. Maintaining Elimination Habits

- Patients normally void on awakening or before meals; therefore, offer the opportunity to use toilet facilities.

- Maintain patients' privacy for voiding. If a patient cannot reach the bathroom and uses a bedside commode or bedpan, make sure that the bedside curtain is closed.

3. Maintaining Adequate Fluid Intake

- Maintaining optimal fluid intake to promote normal micturition.
- A patient with normal renal function who does not have heart or kidney disease needs to take 2200 to 2700 mL of fluid daily.
- Increasing fluid intake helps flush out solutes or particles that collect in the urinary system.

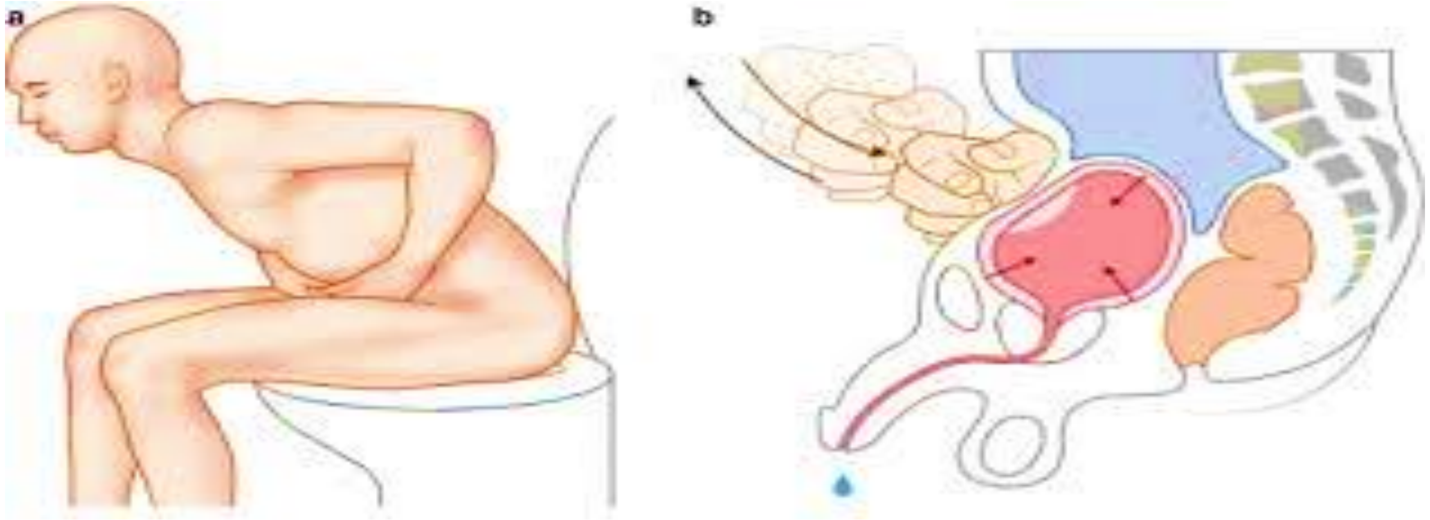
4. Promoting Complete Bladder Emptying

- Under normal conditions a small amount of a patient's urine remains in the bladder after voiding (residual urine).
- Encouraging patients to wait until urine stops flowing or to attempt to void again (double voiding) can improve bladder emptying.

Urinary retention care includes:

1. Scheduled toileting.

2. **Credé's method** or manual compression of the bladder walls with each attempted void may be used. Instruct the patient to place both hands flat on the abdomen below the umbilicus and above the symphysis pubis with the fingers pointed down toward the bladder dome. Have him or her compress the hands downward against the walls of the bladder while tightening the perineum, contracting the abdominal wall, and holding the breath. The maneuver promotes bladder emptying by relaxing the urethral sphincter.



5. Preventing Infection:

Measures to prevent infection of the urinary system:

1. A minimal daily fluid intake of 1200 to 1500 mL dilutes urine.
2. Practice frequent voiding (every 2 to 4 hours) to flush bacteria out of the urethra and prevent organisms from ascending into the bladder.
3. Voiding after intercourse, not using excessive soap or taking bubble baths.
4. Avoid tight-fitting pants or other clothing that creates irritation to the urethra and prevents ventilation of the perineal area.

5. Wear cotton rather than nylon underclothes. Accumulation of perineal moisture facilitates bacterial growth. Cotton enhances ventilation of the perineal area..

6. Drinking enough fluids, especially fluids high in acid ash such as apple or cranberry juice help prevent urinary tract infection (UTI).

7. Girls and women should always wipe the perineal area from front to back following urination or defecation in order to prevent introduction of gastrointestinal bacteria into the urethra.

8. If recurrent urinary infections are a problem, take showers rather than baths. Bacteria present in bath water can readily enter the urethra.

C. Acute Care

1. **Medications:**

- Drug therapy given alone or with other therapies often helps problems of incontinence or retention.
- The bladder is innervated by the parasympathetic nervous system. Drugs that block the muscarinic receptors suppress bladder contractions and reduce incontinence caused by bladder irritation.
- The dribbling or overflow incontinence seen in men with prostatic enlargement can be treated with an alpha1-adrenergic blocker.

2. Catheterization

- Catheterization of the bladder involves introducing a latex or plastic tube through the urethra and into the bladder. The catheter provides a continuous flow of urine in patients unable to control micturition or those with obstructions.

- It also provides a means of assessing urine output in hemodynamically unstable patients. Because bladder catheterization carries the risk of UTI, blockage, and trauma to the urethra, it is preferable to rely on other measures for either specimen collection or management of incontinence.

D. Restorative Care

1. Strengthening Pelvic Floor Muscles

- Patients who have stress or urge urinary incontinence may benefit from **pelvic floor exercises** also known as **Kegel exercises that** improve the strength of pelvic floor muscles (PFM). The client can identify the perineal muscles by tightening the anal sphincter as if to control the passing of gas or to hold a bowel movement.

- A patient begins these exercises during voiding to learn the technique, then practiced at non-voiding times.

2. Bladder Retraining

- Determine the client's voiding pattern and encourage voiding at those times, or establish a regular voiding schedule and help the client to maintain it, whether the client feels the urge or not (e.g., on awakening, every 1 or 2 hours during the day and evening, before retiring at night, every 4 hours at night).

- Regulate fluid intake, particularly during evening hours, to help reduce the need to void during the night.

- Avoid excessive consumption of citrus juices, carbonated beverages (especially those containing artificial sweeteners), alcohol, and drinks containing caffeine because these irritate the bladder, increasing the risk of incontinence.

- Schedule diuretics early in the morning.

3. Measures used to control urination for patients with stress urinary incontinence:

- Learning exercises to strengthen the pelvic floor.
- Initiating a toileting schedule on awakening, at least every 2 hours during the day and evening, before getting into bed, and every 4 hours at night.
- Avoiding an overfilled bladder because this increases chances of incontinence related to increased bladder pressure.
- Minimizing tea, coffee, other caffeine drinks (increases urine production, frequency, and urgency) and alcohol (increases urine production)
- Taking prescribed diuretic medication early in the morning.
- Following a weight-control program if obesity is a problem that is causing increased abdominal pressure.

4. Self-Catheterization

Some patients with chronic disorders such as spinal cord injury learn to perform self-catheterization. The goal is to have patients perform self-catheterization 4 to 6 times a day with volumes of 400 to 500 mL according to the schedule.

5. Maintenance of Skin Integrity

- Washing the skin with mild soap and warm water.
- Perform partial bath and change patient clothes after voiding.
- When the skin becomes irritated or inflamed the health care provider prescribes a cream or spray containing steroids to reduce inflammation.
- Use the antifungal drug that is available in cream or powder if a fungal growth was developed.

6. Promotion of Comfort

- Keep the skin clean, dry and use a protective pad for incontinent patient.

Nursing evaluation

Using the overall goals and desired outcomes identified in the planning stage, the nurse collects data to evaluate the effectiveness of nursing activities. If the desired outcomes are not achieved, explore the reasons before modifying the care plan.


THANK
YOU